

# Supplementary Material

## Can Ambient WiFi Replace LiDAR for Automotive SLAM?

### Localization Yes, Mapping No — and Why

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#### I. REPRODUCIBILITY

Every quantitative claim in the paper is reproduced from a committed artifact. The repository contains the pipeline, the configurations, the result files, and the scripts that generate each figure; a test asserts that the figures are computed *from* the result files, so a figure cannot silently drift from the data behind it.

##### A. Recipe

Install the package and its simulation extra, then run the experiment for the section of interest. Ray-traced runs take the sample count from `WRS_NUM_SAMPLES` (we use  $10^6$  throughout).

TABLE I  
SECTION → SCRIPT → ARTIFACT.

| Section        | Script                                | Artifact                                  |
|----------------|---------------------------------------|-------------------------------------------|
| IV (LiDAR A)   | <code>run_lidar_geo.py</code>         | <code>lidar_geo_results.json</code>       |
| IV (LiDAR B)   | <code>run_lidar_sionna.py</code>      | <code>lidar_sionna_results.json</code>    |
| IV (KITTI)     | <code>run_lidar_kitti.py</code>       | <code>kitti_results.json</code>           |
| V (WiFi)       | <code>regen_wifi_results.py</code>    | <code>wifi_results_paper2.json</code>     |
| VI (cost)      | <code>run_cost_model.py</code>        | <code>cost_results.json</code>            |
| VII (fusion)   | <code>run_fusion.py</code>            | <code>fusion_results.json</code>          |
| VIII (ladder)  | <code>run_enhanced_map.py</code>      | <code>enhanced_map_results.json</code>    |
| VIII (ceiling) | <code>isolate_mapping_floor.py</code> | <code>mapping_floor_isolation.json</code> |
| Figures        | <code>make_paper2_figures.py</code>   | <code>docs/assets/paper2_figures</code>   |

##### B. Stochasticity

The WiFi pipeline is stochastic (ray-tracer sampling, CSI noise, and a particle filter), so WiFi results in the paper are reported as **mean ± standard deviation over five seeds** rather than a single run. Table II gives the per-seed spread for the headline configuration. We recommend that any reproduction quote the spread, not one realization: the seed-to-seed range of the realistic controlled ATE is 0.056–0.143 m, which is wider than several effects one might otherwise be tempted to claim.

TABLE II  
PER-SEED ATE (M), REALISTIC JOINT 2-D MUSIC, CONTROLLED WALL.  
A SINGLE RUN IS NOT A CLAIM.

| Seed    | 42    | 1     | 2     | 3     | 4     | 5     |
|---------|-------|-------|-------|-------|-------|-------|
| ATE (m) | 0.143 | 0.108 | 0.056 | 0.089 | 0.092 | 0.097 |

#### II. THE LiDAR ENVELOPE

Because “LiDAR” is not one sensor, we bracket it. **Model A** slices each scene object at the scan plane and ray-casts the resulting wall segments with occlusion-correct nearest-hit: crisp returns, sparse coverage. **Model B** models the sensor as a monostatic node co-located with the vehicle and gives the scene materials a diffuse scattering coefficient, taking the single-scatter interaction vertices of the ray tracer as returns: dense coverage, noisier points. Both run at mid-range spinning-LiDAR parameters (120 m,  $\pm 3$  cm,  $360^\circ$ ).

A physically important detail: with *specular-only* propagation a monostatic node receives essentially nothing (one retro-reflection in our test), because a specular surface returns energy to the source only at normal incidence. Diffuse backscatter is what makes an EM ray tracer behave like a LiDAR at all — with it, the same scene yields thousands of single-scatter returns. Model B is therefore not merely “LiDAR in Sionna”; the scattering model is the load-bearing choice.

#### III. WHY THE MAPPING CEILING IS NOT PATH DISCRIMINATION

The paper’s isolation experiment matches every MUSIC detection to its nearest true ray-traced path and, for those matching a genuine single-bounce facade path, triangulates the same path twice — once with the true delay and azimuth, once with MUSIC’s estimates. Any degradation is then pure estimation error with discrimination held perfect.

The result is that  $\approx 89\%$  of detections match *no* real path (they are estimator artefacts), a further 2–8% match real but non-facade paths (the discrimination failure that earlier work identified), and only 2–9% match a genuine facade path. Of the last group, on the controlled wall a 6.45 m median *range bias* collapses triangulation success from 100% to 2.4% within 1 m.

Two consequences follow, and they are the reason the paper reports a negative result rather than a method. First, a filter *selects* among detections: it can neither invent the real paths that most detections fail to correspond to, nor correct the bias that ruins the rest. Second, the bias exceeds the 0.94 m resolution limit at 160 MHz by nearly seven times, so it is not a bandwidth problem — consistent with the earlier finding that 60 GHz with 1.76 GHz of bandwidth and sixteen antennas did not move map accuracy.

#### IV. FUSION: THE PARITY CONDITION

Tight fusion carries two independent likelihoods in one particle filter — the WiFi bistatic reprojection and a LiDAR

scan-match — with the output map the union of both sensors’ points. The scan-match target is the accumulated *LiDAR* map only; realistic-CSI WiFi reflectors are several metres from the true surfaces and would corrupt registration if used as a registration target.

The weighting is fixed (equal confidence) and deliberately *not* tuned per scene. That choice is what exposes the paper’s fusion finding: fusion reinforces when the two sensors are of comparable accuracy and *degrades the stronger sensor* when they are badly mismatched (street/LiDAR-A: 0.027 m alone, 0.218 m fused). Had we tuned the weighting per scene, the regression would have disappeared and the condition would have gone unnoticed. Confidence-adaptive weighting is the obvious remedy and is named as future work, not claimed as a result.

## V. COST MODEL

Prices are USD ranges with a source and a date recorded per entry in the released artifact; entries drawn from press reporting carry a verification flag. The model reports ranges throughout and never a point estimate. LiDAR rows in the cost-normalized comparison are priced at the *tier whose parameters we simulate* (mid-range spinning, \$8,000–24,000): crediting the accuracy of that sensor to the price of a budget solid-state unit we never simulated would be an apples-to-oranges comparison. The budget 2-D scanner (\$99) appears as a price *floor*, not as an automotive peer, and we never compare mapping quality against it.